# Naming the Angle Pairs a Transversal Makes
## Answer Key

High-school geometry

### Quick Answers

1. (b) the measure of angle 6 = 62 degrees, —
2. 15
3. 75 degrees
4. 25
5. (b) the measure of angle 4 = 106 degrees, —
6. (a) x = 15, (b) the measure of angle 3 = 70 degrees
7. (a) x = 18, (b) the measure of angle 1 = 85 degrees
8. (b) x = 17, (c) the measure of angle 5 = 106 degrees, —

### What is verified

5 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 5, 8. The worked solution states what a correct response must contain.

### Curriculum

**Level:** High-school geometry
HSG-CO.C – *problems 1, 2, 3, 4, 5, 6, 7, 8*
*Difficulty 1–5 of 5 across 14 tagged checks.*

**1.** $j \parallel k$, eight numbered angles. (a) Name $\angle 3/\angle 6$ and $\angle 4/\angle 6$. (b) $m\angle 4 = 118°$, find $m\angle 6$.

**(a)** $\angle 3$ and $\angle 6$ are **alternate interior angles**: both lie between $j$ and $k$, and they sit on opposite sides of $t$. With $j \parallel k$ they are *congruent*.

$\angle 4$ and $\angle 6$ are **same-side interior angles** (also called co-interior): both lie between $j$ and $k$, and both sit on the same side of $t$. With $j \parallel k$ they are *supplementary*.

**(b)** Same-side interior angles are supplementary, so $m\angle 6 = 180 - 118$. $\boxed{62°}$

**2.** Corresponding angles $(3x + 20)°$ and $(5x - 10)°$; $j \parallel k$. Find $x$.

**Solution:** corresponding angles on parallel lines are congruent, so the two measures are equal.

$$3x + 20 = 5x - 10$$
$$30 = 2x$$

So $x = 15$. Check: both angles measure $3(15) + 20 = 65$ and $5(15) - 10 = 65$. $\boxed{x = 15}$

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**3.** $j \parallel k$, $m\angle 1 = 105°$. Find $m\angle 6$.

**Solution, two links:** $\angle 1$ and $\angle 5$ are corresponding angles (both above their line, both left of $t$), so $m\angle 5 = 105$. Then $\angle 5$ and $\angle 6$ are a linear pair on line $k$, so their measures add to 180: $m\angle 6 = 180 - 105$. $\boxed{75°}$

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**4.** Same-side interior angles $(2x + 30)°$ and $(4x)°$; $j \parallel k$. Find $x$.

**Solution:** same-side interior angles on parallel lines are supplementary — add, do not equate.

$$\begin{aligned}(2x + 30) + 4x &= 180\\ 6x + 30 &= 180\\ 6x &= 150\end{aligned}$$

So $x = 25$. Check: $2(25) + 30 = 80$ and $4(25) = 100$, and $80 + 100 = 180$. $\boxed{x = 25}$

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**5.** No parallel marks. (a) Which relationships survive? (b) $m\angle 2 = 74°$, find $m\angle 4$.

**(a)** The relationships that live at a *single* vertex survive, because they follow from two lines crossing and nothing more: **vertical angles are still congruent** ($\angle 1 \cong \angle 4$, $\angle 2 \cong \angle 3$, and likewise at the lower vertex) and **a linear pair is still supplementary**.

The relationships that connect the *two* vertices all require $j \parallel k$: corresponding, alternate interior, alternate exterior, and same-side interior. Here they carry no guarantee — the two lines meet somewhere off the page, and the angles differ by exactly that much.

**(b)** $\angle 2$ and $\angle 4$ share the ray of $j$ to the right of $t$, and their other sides are the two opposite rays of $t$, so they are a linear pair — which needs no parallel assumption. $m\angle 4 = 180 - 74$. $\boxed{106°}$

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**6.** Alternate interior: $m\angle 3 = (4x + 10)°$, $m\angle 6 = (6x - 20)°$. (a) $x$. (b) $m\angle 3$.

**(a)** Alternate interior angles on parallel lines are congruent:

$$\begin{aligned}4x + 10 &= 6x - 20\\ 30 &= 2x\end{aligned}$$

So $x = 15$. $\boxed{x = 15}$

**(b)** $m\angle 3 = 4(15) + 10 = 60 + 10$. Check: $m\angle 6 = 6(15) - 20 = 70$ too. $\boxed{70°}$

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**7.** Alternate exterior: $m\angle 1 = (5x - 5)°$, $m\angle 8 = (3x + 31)°$. (a) $x$. (b) $m\angle 1$.

**(a)** Alternate exterior angles on parallel lines are congruent — both lie outside the two lines, on opposite sides of $t$:

$$\begin{aligned}5x - 5 &= 3x + 31\\ 2x &= 36\end{aligned}$$

So $x = 18$. $\boxed{x = 18}$

**(b)** $m\angle 1 = 5(18) - 5 = 90 - 5$. Check: $m\angle 8 = 3(18) + 31 = 85$ too. $\boxed{85^\circ}$

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**8.** $m\angle 3 = (4x + 6)^\circ$, $m\angle 5 = (6x + 4)^\circ$, $j \parallel k$. (a) Name the pair. (b) $x$. (c) $m\angle 5$.

**(a)** $\angle 3$ and $\angle 5$ are **same-side interior angles**: both lie between $j$ and $k$, and both are on the left of the transversal. That name means *supplementary*, not congruent, so the two expressions have to be added and the sum set equal to 180. Setting them equal would be the alternate-interior rule applied to a same-side pair — the single most common error on this topic.

**(b)**

$$\begin{aligned}(4x + 6) + (6x + 4) &= 180\\ 10x + 10 &= 180\\ 10x &= 170\end{aligned}$$

So $x = 17$. $\boxed{x = 17}$

**(c)** $m\angle 5 = 6(17) + 4 = 102 + 4$. Check: $m\angle 3 = 4(17) + 6 = 74$, and $74 + 106 = 180$. $\boxed{106^\circ}$